CHAPTER 1

Ratio and Proportion

Ratio

The relationship between two quantities of the same kind is expressed in the form of a ratio. A ratio, therefore is one abstract number divided by another abstract number. For example, the ratio between 7 and 8 is written as 7/8 or 7 : 8. It should be noted that only like things can be expressed as a ratio and not unlike things. For instance we may speak of the ratio of Rs. 4 to 5 is 4/5.

In the ratio a : b, a and b are called the terms of the ratio. a is called the antecedent and b is called the consequent.

It must be noted that a ratio is purely an abstract number and that the two magnitudes must be of the same kind. Further the two terms of the ratio must be expressed in the same unit. A ratio is unaltered if each of its terms is multiplied or divided by the same number.

When two or more ratios are multiplied together, they are said to be compounded.

$A^2 : b^2$ is called the duplicate ratio of a : b.

$A^3 : b^3$ is called the triplicate ratio of a : b.

$\sqrt{a} : \sqrt{b}$ is called the sub-duplicate ratio of a : b.

$3\sqrt{a} : 3\sqrt{b}$ is called the sub-triplicate ratio of a : b.

If a = b, the ratio a : b is a ratio of equality.

If a > b, the ratio a : b is a ratio of greater inequality.

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If $a < b$, the ratio $a : b$ is a ratio of lesser inequality.

Some theorems

Theorem : By adding the same positive number to both it terms, a ratio of greater inequality is diminished and ratio c lesser inequality is increased.

Theorem : By subtracting the same positive number whicl is less than the consequent from both its terms, a ratio of greater inequality is increased and a ratio of lesser inequality is diminished.

Proportion

Four quantities a, b, c, d are said to be *in proportion if* $a : b = c : d$.

a, b, c, d are called proportionals.

a and d are called 'extremes'

b and c are called 'means'

d is called the fourth proportion to a, b, c.

If $a/b = b/c = c/d = d/e = \ldots\ldots$, then the quantities a, b, c, d, e ..... are said to be in continued proportion.

If $a/b = b/c$, then b is called the mean proportional between a and c and c is called the third proportional to a and b.

Theorems

(1) If $a/b = c/d$, then $ad = bc$
and conversely, if $ad = bc$, then $a/b = c/d$

(2) If $a/b = b/c$, then $ac = b^2$
and conversely, if $ac = b^2$, then $a : b = b : c$.

(3) If $a/b = c/d$, then we have the following important results.

(*i*) $b/a = d/c$ (Invertendo)

(*ii*) $a/c = b/d$ (Alternendo)

(*iii*) $(a + b)/b = (c + d)/d$ (Componendo)

(*iv*) $(a - b)/b = (c - d)/d$ (Dividendo)

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(v) $(a + b)/(a - b) = (c + d)/(c - d)$ (Componendo – Dividendo)

Note the following results:

(1) If $a/b = c/d = e/f$, then
each ratio $= (ma + nc + pe)/(mb + nd + pf)$,
where m, n, p are real numbers and $mb + nd + pf \neq 0$.

(2) If $a/b = c/d = e/f = $, then,
each ratio is equal to $[(pa^n + qc^n + re^n + ...)/(pb^n + qd^n + rf^n + ...)]^{(1/n)}$
where p, q, r ..... are any constants, all of them being non-zero.

(3) If a1/b1, a2/b2, a3/b3.....an/bn are all unequal ratios in which the denominators are positive, then the value of $(a1 + a2 + a3 + ... + an)/(b1 + b2 + b3 + ... + bn)$ lies between the least and the greatest of them.

Ex. Aequali

(1) If $a/b = x/y$ and $b/c = y/z$ then $a/c = x/z$

(2) $a : b : c = x : y : z$ denotes that $a/b = x/y$, $b/c = y/z$ and $a/c = x/z$
also, $a : b : c = x : y : z$ implies $a/x = b/y = c/z$

Ratio & Proportion

Ex. 1. A fertilizer contains 4 parts of Phosphate, 6 parts Nitrogen and 5 parts Potash. Find the number of Kilograms of fertilizer that will contain 80 Kilograms of Phosphate.

Sol. : The ratio of Kilos of Phosphate to Kilos of fertilizer $= 4/(4 + 6 + 5) = 4/15$
Suppose x is the number of Kilograms of the mixture fertilizer then,

$4/15 = 80/x$

$x = (80 \times 15)/4 = 300$ Kilos of fertilizer.

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Ex. 2. In a village of 2800 persons, 1600 are females, find the ratio of males to females in that village.

Sol. : males = 2800 – 1600

$\therefore$ males/females = (2800 – 1600)/1600

= 1200/1600 = ¾

Express this ratio as a percentage :

Here, ¾ × 100 = 75%

Express the ratio as a decimal fraction :

Here, $^3/_4 = 0.75$

Ex. 3. The incomes of A and B are in the ratio 7 : 4 and their expenditures are in the ratio 9 : 5. If each saves Rs. 500, find their respective incomes.

Sol. : Let the respective incomes of A and B be x and y, then

$x/y = 7/4$ ..... (1)

$(x - 500)/(y - 500) = 9/5$ ..... (2)

From equation (1)

$4x = 7y$

$\therefore x = 7/4\ y$

Substituting this value in equation (2), we get

$[(7/4\ y) - 500]/(y - 500) = 9/5$

$5(7/4)y - 2500 = 9y - 4500$

$\therefore$ $35/4\ y - 9y = 2500 - 4500$

i.e. $9y - 35/4\ y = 4500 - 2500$

i.e. $(36y - 35y)/4 = 2000$

i.e. $y/4 = 2000$

$\therefore$ $y = 8000$

$x = 7/4\ y = 7/4 \times 8000 = 14000$

$\therefore$ The incomes of A and B are Rs. 14000 and Rs. 8000 respectively.

Ex. 4. The ratio of the present ages of a man and his wife is 3 : 2. Find their present ages if five years hence the man would be twice as old as his wife was five years back.

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Sol. : Let the ages of the man and his wife be x and y respectively, then

$x/y = 3/2$ ..... (1)

$(x + 5)/(y - 5) = 2$ ..... (2)

$x = 3/2\ y$ from equation (1),

substituting this value in equation (2), we get

$(3/2\ y + 5)/(y - 5) = 2$

$3/2\ y + 5 = 2(y - 5)$

$\therefore\ 5 + 10 = 2y - 3/2\ y$

$\therefore\ 15 = ½\ y$

$\therefore\ y = 30$

$x = 3/2\ y = 3/2 \times 30 = 45.$

$\therefore$ The present ages of the man and his wife are 45 and 30 years respectively.

Ex. 5. On a council the ratio of men to women was 5 : 2. When the number of members in the council was increased by 4 men and one woman, the ratio changed to 8 : 3. Find the number of members in the council originally.

Sol. : Let the number of men originally be x and the number of women y, then

$x/y = 5/2$ ..... (1)

and $(x + 4)/(y + 1) = 8/3$ ..... (2)

from (1) $2x = 5y$

i.e. $x = 5/2\ y$

$\therefore\ (5/2\ y + 4)/(y + 1) = 8/3$

i.e. $(5y + 8)/2(y + 1) = 8/3$

$\therefore\ 3(5y + 8) = 16(y + 1)$

$15y + 24 = 16y + 16$

$24 - 16 = 16y - 15y$

$\therefore\ y = 8$

$x = 5/2\ y = 5/2 \times 8 = 20$

$\therefore$ the number of members in the council was 28 originally.

Ex. 6. In a book there are 500 pages and each page

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contains 60 lines. How many lines should be there on one page, if the whole book is to be completed in 300 pages.

Sol. : Let the number of lines = x then,

$500 \times 60 = 300 \times x$

$\therefore \quad x = 500 \times 60/300 = 100$

$\therefore$ there should be 100 lines on each page if the book is to be completed in 300 pages.

Ex. 7. If 9 men can clear a field of 180 acres in 30 days, what would be the area of the field which can be cleared by 25 men in 45 days?

Sol. : Suppose the area of the concerned field is x then,

$25/9 \times 45/30 = x/180$

$\therefore \quad x = 180 \times 25/9 \times 45/30 = 750$ acres.

Ex. 8. A person's initial salary is Rs. 3000 p.m. If he gets yearly increment of Rs. 60, find the ratio of his salary after 5 years to his initial salary.

Sol. : After 5 years the person's salary would be $3000 + 300 = 3300$

$\therefore$ the required ratio = 3300/3000 = 11/10

Ex. 9. The prices of three articles are in the ratio 2 : 3 : 4. If the price of the 2nd article is Rs. 96, find the prices of the 1st and 3rd article respectively.

Sol. : Let the prices of the 1st and 3rd article be x and z then

$x/2 = 96/3 = z/4$

i.e. $x/2 = 32$ and $z/4 = 32$

$\therefore \quad x = 64$ and $z = 128$

$\therefore$ the 1st and the 3rd articles are priced at Rs. 64 and Rs. 128.

Ex. 10. X is 32 years old and Y is 21 years old. What is the least number of years after which the ratio of the age of X to that of Y will be less than 9 : 7?

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Sol. : Let the 'least' number of years be k, then

$(32 + k)/(21 + k) = 9/7$

i.e. $224 + 7k = 189 + 9k$

$224 - 189 = 2k$

$35 = 2k$

$\therefore \quad k = 17\frac{1}{2}$

$\therefore$ At least 18 years must pass so that the ratio of the age of X to that of Y may be less than 9 : 7

Ex. 11. In 50 Kilograms of a mixture of milk and water, there is 10% water. Find the quantity of water that should be added so as to increase the percentage of water to 25%.

Sol. : Mixture of 50 kgs. contains milk 90% and water 10%

Quantity of water in the mixture

$= (50 \times 10)/100 = 5$ kg.

$\therefore$ Quantity of milk in the mixture = 45 kg.

In the new mixture milk should be 75% and water 25%,

Total qty. of new mixture

$= (100 \times 45)/75 = 60$ kgs

$\therefore$ Quantity of water to be added

$= 60 - 50 = 10$ kgs.

Ex. 12. 72.90 kgs. of a mixture contains milk and water in the ratio 7 : 2. How much more water must be added to this mixture so that the ratio of milk and water may be 7 : 3?

Sol. : Quantity of milk in 72.9 kgs of mixture

$= 72.9 \times 7/9 = 56.7$ kgs.

Quantity of water in that mixture

$= 72.9 \times 2/9 = 16.2$ kgs.

The new mixture is required to contain 56.7 kgs. of milk and water in the ratio 7 : 3

Let the quantity of water be x, then

$56.7/x = 7/3$

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$\therefore \quad x = 56.7 \times 3/7 = 24.3$ kgs.

The quantity of water that must be added to the mixture is

$24.3 \text{ kgs} - 16.2 \text{ kgs} = 8.1 \text{ kgs.}$

Ex. 13. It took 81 men working 8 hours a day to construct a wall 320 yards long, 27 feet high and 6 feet wide. How many men should be engaged to construct a wall 450 yards long, 30 feet high and 8 feet wide, if they are to work 9 hours a day and to finish the work in the same time as the first?

Sol. : Volume of the first wall $= 320 \times 27/3 \times 6/3$
$= 5760$ cu. yds.

Volume of the second wall $= 450 \times 30/3 \times 8/3$
$= 12000$ cu. yds.

Let the number of men to be engaged be x, then

Hours	Work	Men
8	5,760	81
9	12,000	x

$\therefore \quad x = (81 \times 8 \times 12000)/(9 \times 5760) = 150$ men.

Ans. 150 men should be engaged.

Note : If the hours of work per day are more, then less men. If the amount of work is more, then more men.

Ex. 14. A besieged town has provisions to last for 3 weeks, if the population is 22,400. How many people must be sent away in order that the provisions may last for 7 weeks?

Sol. : 22,400 persons have provisions for 3 weeks. Therefore $3 \times 22,400$ persons have provisions for 1 week. If the provisions should last for 7 weeks, the provisions would be adequate for

$(3 \times 22,400)/7 = 9,600$ persons

The number of persons to be sent away is

$22,400 - 9,600 = 12,800$

Ex. 15. A garrison of 1,500 soldiers has provisions for 48 days. At the end of 13 days, a reinforcement arrives

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and it is found that now the provisions will last for 25 days. What is the reinforcement?

Sol. : For 35 days i.e. 48 – 13 days, 1500 soldiers have provisions.

For 25 days $(1500 \times 35)/25 = 2100$ soldiers have provisions.

$\therefore$ The reinforcement consists of

$2100 - 1500 = 600$ soldiers.

Ex. 16. A garrison of 15,000 men is provisioned for 5 months at the rate of 900 grams per man. At the end of two months it was decided to reduce the diet allowance so that the remaining stock may last for 9 months. If in the meanwhile the number of men is reduced to 9,000, what should be the estimated daily allowance per head?

Sol. : At the end of two months, the remaining stock would last for 3 months at 900 grams per man. The daily allowance for 900 men for 9 months is calculated as follows:

Men	Months	Diet
15,000	3	900
9,000	9	x

$\therefore$ $x = 900 \times 15000/9000 \times 3/9 = 500$ grams.

Daily allowance per head is 500 grams.

Note : It should be noted that less men implies more quantity of diet and that more months implies less diet for each.

Ex. 17. According to the terms of a contract, a contractor had to build a house in 60 days, for which he engaged 45 men. After 40 days he found that the work was getting delayed and therefore he engaged 30 additional men. He finished the job 5 days ahead of the schedule. How many days behind time he would have been if he did not employ additional men?

Sol. : 45 men can complete in 60 days whole work

45 men can complete in 1 day, 1/60 work

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45 men can complete in 40 days, 40/60 work.
The work that remained = 1 – 40/60 = 20/60 = 1/3
The work was completed 5 days earlier, i.e. by engaging 30 additional men the work was completed in 15 days. This means that 75 men completed the remaining 1/3 work in 15 days.

∴ 1 man can complete 1/3 work in 75 × 15 days

∴ 45 men can complete 1/3 work in (75/15)/45 = 25 days

∴ The total time taken by 45 men to complete the work would have been 40 + 25 = 65 days.

Therefore if no additional men were engaged, the work would have been completed 5 days later.

Mixtures

When two or more qualities or kinds of the same commodity such as tea, coffee or liquids such as milk, wine, etc. are suitably mixed, the worth of the resulting mixture is calculated in the following manner :—

Suppose,

P1 = price per unit of the quality 'A'
P2 = price per unit of the quality 'B'
P3 = price per unit of the quality 'C'
And w1 = quantity of the quality A
w2 = quantity of the quality B
w3 = quantity of the quality C

then the cost price of the mixture per unit is :

$$Pm = (p1w1 + p2w2 + p3w3 + ...)/(w1 + w2 + w3 + ...)$$

In particular, if two qualities A and B of a commodity are mixed then the cost price or the worth of the mixture per unit is :

$$P = (p1w1 + p2w2)/(w1 + w2)$$

From the above result, we can derive the ratio of the two qualities A and B in the mixture, in the following manner :—

$$p(w1 + w2) = p1w1 + p2w2$$

i.e. $pw1 + pw2 = p1w1 + p2w2$

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$pw1 - p1w1 = p2w2 - pw2$

$w1(p - p1) = w2(p2 - p)$

i.e. $w1/w2 = (p2 - p)/(p - p1)$

In words, the above proportion is :

Quantity of A/Quantity of B = (Price of B – Price of M)/(Price of M – Price of A)

Where price means price per unit and M stands for the mixture of A and B

In case of problems concerning liquids, we take into consideration the strengths of the respective liquids and the resulting mixture before arriving at the ratio of mixture. That is :

Quantity of Liquid L1/ Quantity of Liquid L2
= (Strength of L2 – Strength of M)/ (Strength of M – Strength of L2)

Ex. 18. A grocer mixes 12 kilograms of tea at Rs. 9 per kg. with 8 kilograms of tea at Rs. 11 per kg. At what price per kilogram must he sell the mixture so as to gain 30 paise per kilogram?

Sol. : Cost Price of the Mixture

$= (12 \times 9 + 8 \times 11)/(12 + 8)$

$= (108 + 88)/20$

$= 196/20$

$= 9.80$

Selling Price of the Mixture is 9.80 + 0.30 that is, Rs. 10.10 ps. per kilogram.

Ex. 19. A superior quality of oil at Rs. 6 per kilogram is mixed with an inferior quality of oil at Rs. 4 per kg. If the mixture is worth Rs. 4.50 per kg., in what ratio are the two qualities of oil mixed?

Sol. : Let w1, w2 be the respective weights of the two qualities of oil that are mixed, then the cost price of the mixture is

$(6w1 + 4w2)/(w1 + w2)$

It is given that the mixture is worth Rs. 4.50, therefore,

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$$4.50 = (6w1 + 4w2)/(w1 + w2)$$

i.e. $9/2 = (6w1 + 4w2)/(w1 + w2)$

$$9(w1 + w2) = 2(6w1 + 4w2)$$

$$9w1 + 9w2 = 12w1 + 8w2$$

$$9w2 - 8w2 = 12w1 - 9w1$$

$$\therefore \quad w2 = 3w1$$

$$\therefore \quad w1/w2 = 1/3$$

Alternatively the above ratio can be simply derived at once :

$$w1/w2 = (4 - 4.50)/(4.50 - 6.00)$$
$$= 0.50/1.50 = 1/3$$

Thus the two qualities are mixed in the ratio 1 : 3

Ex. 20. A coffee dealer blends 15 kgs. of coffee at Rs. 12 per kg with 10 kgs. of coffee at Rs. 15 per kg. What is the lowest price per kg. at which he can sell the mixture so as to gain at least 25 p.c.?

Sol. : Cost of 15 kgs. @ Rs. 12 per kg. = 180

Cost of 10 kgs. @ Rs.15 per kg. = 150

$\therefore$ Cost of 25 kgs. of mixture = 330

25% of $330 = 330 \times 25/100 = 82.50$

To make a gain of 25 p.c. the dealer should sell 25 kgs. of mixture for Rs. 330 plus 82.50, i.e. for Rs. 412.50

$\therefore$ S.P. per kg. of mixture

$= 412.50/25 =$ Rs. 16.50.

Ex. 21. Two vessels contain mixtures of milk and water. In one vessel milk and water are in the ratio of 5 : 3 and in the other in the ratio 3 : 1. In what proportion the quantities from the two vessels should be mixed together so that the mixture formed may contain milk and water in the ratio 7 : 3?

Sol. : Let S1 be the strength of the mixture (milk + water) in the first vessel and S2 the strength of the mixture in the second vessel, and Sm the strength of the new

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mixture formed in another vessel, then the strength of milk :

$S1 = 5 : 8$, $S2 = 3 : 4$ and $Sm = 7 : 10$

Therefore,

Quantity of mixture from the first vessel/ Quantity of mixture from the second vessel

$= (S2 - Sm)/(Sm - S1)$

$= (¾ - 7/10)/(7/10 - 5/8)$

$= [(30 - 28)/40]/[(56 - 50)/80]$

$= 2/40 \times 80/6$

$= 4/6 = 2/3$

Ex. 22. A coffee dealer sells one kind of coffee at Rs. 8.50 per kg and loses 15 p.c. and another at Rs. 15 per kg. and gains 20 p.c. How should the two qualities of coffee be mixed so that the blended coffee could be sold at 14 per kg. at a profit of 25 p.c.?

Sol. : At 85 p.c., the c.p. of the first kind of coffee is :

$85/100 \times C.P = 8.50$

$C.P = 8.50 \times 100/85 = 850/85 = 10$

At 120 p.c, the C.P. of the second kind of coffee is :

$120/100 \times C.P = 15$

$C.P = 15 \times 100/120 = 12.50$ Ps.

At 125 p.c., the cost of the blended coffee is :

$125/100 \times C.P = 14$

$C.P = 14 \times 100/125 = 14 \times 4/5$

$= 56/5 = 11.20$

The required ratio for blending the two qualities of coffee is :

Qty of First Quality/Qty of Second Quality

= (Cost of the Second Quality – Cost of Blend)/ (Cost of Blend – Cost of First Quality)

$= (12.50 - 11.20)/(11.20 - 10.00)$

$= 1.30/1.20$

$= 13/12$

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